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DEPARTMENT OF CSE/CS&IT

PROJECT ABSTRACT SUBMISSION

COURSE: Cloud Based Data Analytics System, 23SDCS08 A.Y. 2025 - 2026

TITLE OF THE PROJECT	TRAVEL BEHAVIOUR ANALYTICS AND SMART TRIP RECOMMENDATION SYSTEM	
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Abstract

In today's digital era, large volumes of travel data are generated through online booking systems and tourism platforms. Analyzing this data is essential for understanding travel patterns, cost distribution, and customer behavior. This project, titled "Travel Data Analysis and Recommendation System using Azure Machine Learning and Power BI," focuses on extracting meaningful insights from travel data using cloud-based technologies and visualization tools.

The project utilizes Microsoft Azure Machine Learning for data handling and model training, and Power BI for data visualization. The dataset includes attributes such as destination, travel duration, accommodation type, accommodation cost, transportation cost, and traveler details. During preprocessing, data cleaning techniques were applied to remove unnecessary symbols, handle missing values, and convert data into appropriate formats.

Automated Machine Learning (AutoML) was used to train models, and the best-performing model was identified as a Voting Ensemble model. Power BI was used to create interactive visualizations such as bar charts, pie charts, and maps to represent travel trends and cost analysis effectively.

The results highlight key insights such as peak travel months, most expensive destinations, and distribution of accommodation types. The project demonstrates that cloud-based analytics combined with visualization tools can efficiently process and analyze large datasets, supporting better decision-making.

Overall, this system provides a scalable and efficient approach to travel data analysis and can be further extended for real-time applications and recommendation systems.